Lecture 31: Materials & Radiation Damage

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

April 1, 2026

1 Introduction

So far, we have treated the reactor core as a static geometry. In reality, the core operates in an extreme environment. Materials are subjected to:

1. **Intense Radiation:** High flux of neutrons ($10^{14}\, n/cm^2 \cdot s$) and gammas.
2. **High Temperature:** Gradients create thermal stress.
3. **Corrosive Chemistry:** High-temperature water, boric acid, or liquid sodium.

This lecture focuses on **Radiation Damage**: how neutrons physically alter the atomic structure of materials, leading to failure modes like embrittlement, swelling, and gas buildup.

2 Mechanisms of Radiation Damage

When a fast neutron enters a solid lattice, it transfers significant kinetic energy to the stationary atoms, initiating a series of scattering events.

2.1 Atomic Displacement

- **Primary Knock-on Atom (PKA):** The first atom struck by the neutron. It recoils with high kinetic energy (keV range).
- **Displacement Cascade:** The PKA travels through the lattice, initiating a cascade of secondary and tertiary collisions. A single 1 MeV neutron can displace $\approx 500 - 1000$ atoms.
- **Frenkel Defects:** The result of these collisions is a pair of point defects:
  1. **Vacancy:** An empty lattice site (hole).
  2. **Interstitial:** An atom forced into a non-lattice position.

2.2 Quantifying Damage: dpa

We measure damage not in "years," but in **Displacements Per Atom (dpa)**.

$$1 \text{ dpa} = \text{Every atom in the lattice has been knocked out of place once.}$$

- *Scale:* The extent of damage depends heavily on the component's proximity to the fuel:

- **Core Internals (Stainless Steel):** Components directly surrounding the fuel (baffle plates, bolts, core barrel) might see **50–100 dpa** over a 60-year lifetime. At this level, the material's atomic structure is highly dynamic, continually breaking and reforming its crystal lattice.
- **Reactor Pressure Vessel (Ferritic Steel):** The massive outer vessel is shielded by the internal components and the water coolant, seeing only **0.01 to 0.1 dpa** over its life. However, even this comparatively tiny dose is enough to cause severe structural issues (embrittlement) over time.

3 Macroscopic Effects of Damage

3.1 Radiation Hardening and Embrittlement

Defects (interstitials/vacancies) cluster together to form "dislocation loops." These impede the movement of slip planes in the metal.

- **Hardening:** The metal becomes harder and stronger (yield stress increases).
- **Embrittlement:** The metal loses ductility. Instead of stretching under stress, it fails via brittle fracture.

3.1.1 The DBTT Shift (Crucial for Pressure Vessels)

Ferritic steels (carbon steel used in vessels) undergo a transition from Ductile (tough) to Brittle (glass-like) as temperature drops.

- **DBTT:** Ductile-to-Brittle Transition Temperature.
- **Effect of Flux:** Neutron irradiation shifts the DBTT to *higher* temperatures.
- **Safety Consequence (Pressurized Thermal Shock - PTS):** If the vessel's DBTT rises from $-20°C$ to $+100°C$ due to age:
  - Imagine a LOCA (Loss of Coolant Accident) occurs.
  - Emergency Core Cooling Systems (ECCS) inject cold water (20°C) into the hot vessel.
  - If $T_{water} <$ DBTT, the vessel is in the **Brittle** regime. The thermal shock could induce rapid crack propagation and brittle fracture within the vessel wall.

3.2 Void Swelling (Volumetric Expansion)

While embrittlement changes the *properties* of the metal, swelling changes its *dimensions*.

- **Mechanism:** At high temperatures ($> 300°C$) and high flux, vacancies created by radiation can migrate and cluster together to form empty 3D holes called "voids."
- **Effect:** The creation of these internal voids reduces the density of the metal, causing the bulk material to expand physically.
- **Relevance:**
  - *LWRs:* A minor issue for most components, but significant for "Core Internals" (baffle bolts and plates) which can distort over 60+ years.

  - *Fast Reactors:* A primary design constraint. Stainless steel cladding in Fast Breeder Reactors can swell by > 10% by volume, causing fuel assemblies to bow and jam.

3.3 Irradiation Creep

Creep is the slow, time-dependent deformation of a material under constant stress.

- **Thermal Creep:** Occurs at high temperatures (typically $> 0.4\,T_{melt}$) where atoms possess sufficient thermal energy to diffuse and relieve stress.
- **Irradiation Creep:** Occurs at *much lower* temperatures in a reactor core. The constant kinetic displacement of atoms (dpa) provides the mobility for the metal to slowly flow and deform under applied stress (like coolant pressure), even when thermal creep would otherwise be negligible.

4 Fuel Rod Mechanics: Gas and Pressure

The fuel rod is not just a simple receptacle for uranium; it acts as a primary pressure boundary in its own right.

4.1 Fission Gas Release

Uranium fission produces noble gases (Xenon and Krypton) as direct fission products.

- **Insolubility:** Being noble gases, they do not chemically bond or dissolve in the UO_2 crystal lattice.
- **Migration:** At high temperatures (> 1000°C), these gas atoms migrate to grain boundaries, form bubbles, and tunnel their way out of the pellet into the "Gap" (the space between fuel and cladding).
- **The Feedback Loop:**
  1. The rod is initially filled with Helium (excellent thermal conductivity).
  2. As Xe/Kr (low thermal conductivity) mix in, the "Gap Conductance" drops.
  3. This decreases heat transfer, making the fuel hotter → causing *more* gas release → causing *lower* conductance.

4.2 Pressure Balancing and "Compaction" (Pre-Pressurization)

The cladding tube (Zircaloy) is thin (≈ 0.6 mm). It faces a substantial pressure differential:

- **External Pressure:** In a PWR, the coolant is at ≈ 2250 psi (15.5 MPa).
- **Internal Pressure:** If the rod were filled with air at 1 atm, the external water pressure would cause rapid inward deformation of the cladding. This is called **Cladding Creep Collapse** (flattening the tube onto the fuel).

The Solution (Pre-Pressurization): We manufacture the rods by pressurizing them with Helium to $\approx 300 - 400$ psi (2-3 MPa).

- This "support pressure" reduces the differential stress (ΔP) on the cladding wall early in life.
- As fission gas builds up later in life, the internal pressure rises, eventually equaling or exceeding the coolant pressure.

4.3 Pellet-Cladding Interaction (PCI)

- **Swelling:** As the fuel burns, it swells (due to solid fission products and gas bubbles).
- **Creep Down:** Simultaneously, the cladding creeps inward due to coolant pressure (driven by the irradiation creep mechanism).
- **The Collision:** Eventually, the gap closes. The fuel pellet physically pushes against the cladding. If power is ramped up too fast, the brittle cladding can crack from the tensile stress exerted by the expanding pellet.

4.4 The Modern Solution: TRISO Fuel

To solve the issues of gas release, swelling, and PCI, advanced reactor designs (and many microreactors) are shifting away from Zircaloy tubes to **TRISO (TRi-structural ISOtropic)** fuel particles.

- **Structure:** A microscopic uranium core is encapsulated in concentric layers of porous carbon (to absorb fission gas swelling), dense pyrolytic carbon, and Silicon Carbide (SiC).
- **Benefit:** Each millimeter-sized particle acts as its own individual, incredibly strong pressure vessel. They can withstand temperatures up to 1600°C without melting or releasing fission products, inherently solving the PCI and gas pressure limits of traditional cladding.

5 Material Case Studies

5.1 Beryllium: A High-Performance Reflector with Drawbacks

Beryllium is neutronically excellent:

- **Pro:** Low mass (good moderator) and creates neutrons via $^{9}\text{Be}(n, 2n)2\alpha$.

Why don't we use it everywhere?

- **Helium Production:** The (n, α) reaction creates Helium atoms inside the metal lattice.

  $$^{9}\text{Be} + n \rightarrow 2\ n + 2\ ^{4}\text{He}$$

  Helium is a noble gas; it does not dissolve. It forms high-pressure gas bubbles at grain boundaries, causing severe **swelling and cracking**.

5.2 Zirconium Alloys (Zircaloy)

Used for fuel cladding.

- **Composition:** Primarily **Tin (Sn)** ($\approx 1.5\%$) plus Iron/Chromium. Natural Zirconium must have the **Hafnium** (a strong neutron absorber) chemically removed.
- **Con 1: Hydride Embrittlement.** Zirconium reacts with water: $Zr + 2H_2O \rightarrow ZrO_2 + 2H_2$. Some Hydrogen diffuses *into* the metal. Zr acts as a "sink" for hydrogen. The H precipitates as brittle Zirconium Hydride platelets, weakening the tube.
- **Con 2: Rapid Oxidation.** Above 1000°C (accident conditions), the water reaction becomes exothermic and self-sustaining, producing explosive Hydrogen gas (e.g., Fukushima Daiichi).

5.3 Graphite (The Wigner Effect)

- **Mechanism:** Neutrons knock carbon atoms into interstitial sites. These atoms possess high potential energy (stored energy).
- **Release:** If the graphite is heated slightly, these atoms can be triggered to snap back into vacancies all at once.
- **Consequence:** A substantial, rapid release of heat (Wigner Energy Release).

Historical Context: The Windscale Fire (1957)

In October 1957, the UK's Windscale Unit 1 nuclear reactor experienced a severe accident directly linked to the Wigner effect. The graphite moderator had accumulated significant stored energy over its operation. Operators deliberately heated the reactor to induce a controlled release of this energy (a process called annealing). However, the energy release was more rapid and localized than anticipated, leading to extreme temperatures that ignited the graphite moderator and the uranium fuel. The fire burned for three days, releasing substantial amounts of radioactive fallout (particularly Iodine-131) into the environment. It remains one of the worst nuclear accidents in history and serves as a primary case study for the rigorous management of radiation damage in reactor materials.

5.4 Carbon Steel and Boric Acid: The Davis-Besse Incident (2002)

A prime example of **Corrosive Chemistry** interacting with stress and operational degradation.

- **The Mechanism (EAC):** Environmentally Assisted Cracking (EAC) caused microscopic through-wall cracks in the Control Rod Drive Mechanism (CRDM) nozzles penetrating the reactor pressure vessel head.
- **The Consequence:** High-pressure reactor coolant (water laced with boric acid to control reactivity) leaked onto the top of the vessel. As the water flashed to steam, it left behind highly concentrated boric acid.
- **The Damage:** Over months, the acid ate a pineapple-sized hole entirely through the six-inch thick carbon steel vessel head. Only a thin ($\approx 3/8$ inch) layer of internal stainless steel cladding prevented a massive Loss of Coolant Accident (LOCA). It serves as a stark reminder of the destructive potential of primary coolant chemistry.

6 Relevance: Life Extension and Restarts

Materials science is the limiting factor in extending the life of the nuclear fleet.

6.1 The Palisades Restart (Michigan)

In a historic first, the decommissioned Palisades plant is being prepared for restart.

- **The Issue:** The reactor pressure vessel is considered one of the most embrittled in the US fleet (due to alloy chemistry and high fluence).
- **The Cure (Analysis vs. Annealing):**

  - *Annealing:* Physically heating the vessel to 450°C to "heal" defects. This was considered by previous owners decades ago but **never performed**. It is **not** currently planned for the restart.
  - *Advanced Analytical Modeling:* Holtec is relying on **Alternate PTS Rules (10 CFR 50.61a)**. This involves using updated, higher-fidelity fracture mechanics models to demonstrate that the vessel still has sufficient safety margin ("Equivalent Margins Analysis") without physical repair.

- **Controversy:** Critics argue this relies on analytical models while circumventing the physical reality of the degraded steel.

Status Update: Palisades Restart (Spring 2026)

While Palisades successfully transitioned back to "operational status" under the NRC in late 2025 and received its first delivery of new nuclear fuel, the historic restart is currently behind schedule. Originally targeted to resume power generation by the end of 2025, owner Holtec International has pushed the timeline to **Spring 2026** to complete complex maintenance, including the sleeving of thousands of steam generator tubes.
Current Status & Hurdles:

- **Regulatory Scrutiny:** The NRC is maintaining strict oversight, requiring additional detailed engineering analyses regarding aging reactor vessel welds and historical documentation before granting final restart approvals.
- **Legal Opposition:** A coalition of environmental groups filed a federal lawsuit in late 2025 attempting to halt the restart, citing safety concerns over the 50-year-old plant's degraded components.
- **Future Expansion (SMRs):** Despite delays with the legacy reactor, Holtec is advancing plans to expand the site. With Department of Energy backing, early site preparation is slated to begin for two new SMR-300 small modular reactors adjacent to the existing plant once the legacy reactor is safely connected to the grid.

6.2 The Three Mile Island Unit 1 Restart (Economics vs. Materials)

In contrast to Palisades, Constellation Energy is preparing to restart TMI Unit 1 (shut down in 2019, now renamed the Crane Clean Energy Center) by 2028, fueled entirely by a 20-year Power Purchase Agreement with Microsoft.

- **The Contrast:** TMI-1 was shut down prematurely due to poor economics (the inability to compete with cheap natural gas), *not* physical degradation. Its reactor vessel and steam generators remain in remarkably good condition for their age.
- **The Lesson:** The longevity of the nuclear fleet is governed by a split between material health and financial viability. While Palisades is fighting the structural limits of materials science, TMI-1 is primarily fighting a regulatory and non-nuclear physical refurbishment battle.

6.3 Reuse of Naval Reactors (Data Centers)

Recently, proposals have emerged to repurpose decommissioned military reactors for civilian use. Notably, **HGP Intelligent Energy LLC** has proposed extracting the two retired A4W pressurized water reactors from the USS *Nimitz* to power a dedicated AI data center campus.

- **The Logic:** These are highly reliable, self-contained power plants (producing ≈ 500 MWth each) capable of providing decades of continuous, carbon-free baseload power.
- **The Material & Engineering Problems:**
  - *Design Life:* Naval reactors are over-engineered for shock resistance but have rigorous "end of life" limits based on thermal and physical fatigue cycles over a 50-year operational lifespan.
  - *Fuel:* They use weapons-grade **Highly Enriched Uranium (HEU)**, presenting extreme civilian regulatory and proliferation hurdles.
  - *Integrity & Extraction:* These reactors are not modular and were never designed to be easily disassembled. *Nimitz*-class carriers undergo only a single mid-life Refueling and Complex Overhaul (RCOH). The reactor vessel and its heavy shielding are permanently welded into the ship's hull structure to withstand combat damage. Extracting them intact and re-qualifying aged, highly-irradiated naval steel for civilian land-based standards represents a formidable engineering and regulatory challenge.

Discussion Question: Analytical Safety vs. Physical Safety

The Palisades restart relies on "Analytical Safety"—using advanced computer models (10 CFR 50.61a) to prove the reactor vessel is safe despite high embrittlement, rather than "Physical Safety" measures like annealing (which costs $500M+).

- *Prompt:* Is it ethical to substitute higher-fidelity math for physical remediation when public safety is at stake? Does this represent a triumph of engineering understanding (precision) or a normalization of risk (cutting corners) to save a financial asset?

References

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